DossierGraph: AI pre-scrutiny for AICTE approval dossiers

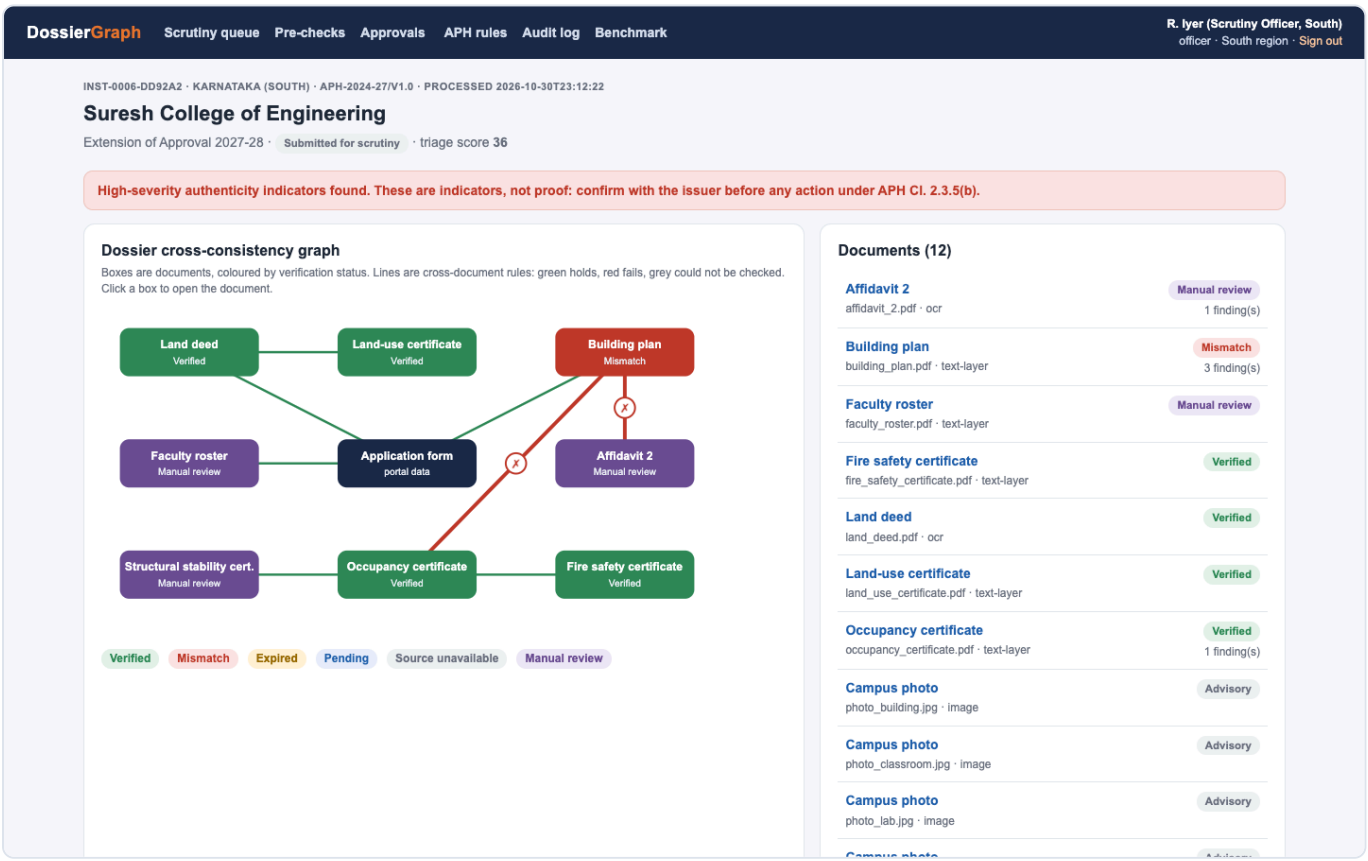
SIH 2026 · Problem Statement SIH26261 · *AI-Powered Application Validation, Document Authentication and Regulatory Compliance Verification* · Team **CITADEL**

DossierGraph checks an institution's whole approval dossier before a human scrutiny officer reads it:

- every document against its **issuing authority** (digital signature, QR code, issuer registry);
- every document against the **AICTE Approval Process Handbook (APH) 2024-27**, with 20 rules that each cite their clause;
- every document against **every other document**: land deed ↔ land-use certificate ↔ building plan ↔ occupancy certificate ↔ fire certificate ↔ Affidavit 2 ↔ faculty roster.

Officers get a ranked queue and a findings list in which every finding is linked to evidence. **The system never decides.** Adverse outcomes need a second, senior officer to approve them.

> **Status:** working prototype, measured on 200 labelled synthetic dossiers (2,281 documents). All data is synthetic; no real institution, person or document is used (see Test documents).



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Quick start

You need **macOS or Linux, Python 3.11 to 3.13 and Tesseract OCR**. On a Mac: `brew install tesseract`; on Ubuntu: `sudo apt install tesseract-ocr`.

```
cd prototype
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
python -m dossiergraph.cli seed          # makes 16 demo dossiers, test users and processes them
(~50 s)
uvicorn dossiergraph.app:app --port 8000
```

Open http://localhost:8000. The test accounts are in `prototype/seed_users.yaml` and are local only:

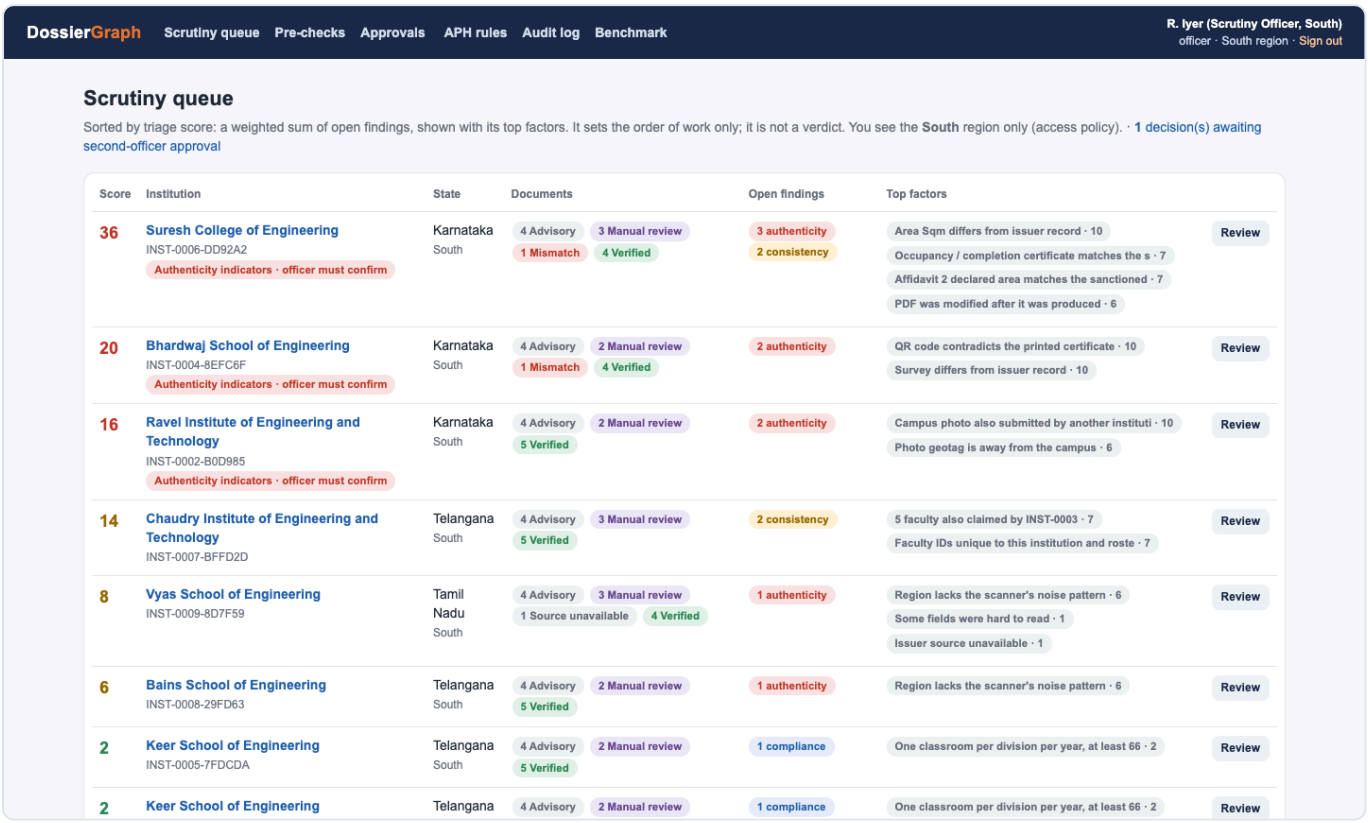
Account	Role	Sees
`officer.south` / `officer.west` / `officer.north`	Scrutiny officer	Only their region's applications
`senior`	Senior officer	All regions; the only role that approves adverse outcomes
`admin`	Administrator	Rules, audit log, benchmark; cannot decide applications
`inst-0001` … `inst-0016`	Institution	Only its own applications and its own access log

Other commands:

```
python -m pytest -q                                  # 14 tests, ~30 s
python -m bench.run_benchmark --n 200 --seed 2026    # full benchmark, ~7 min
python -m synth.generate --set mine --n 50 --seed 7  # generate more labelled dossiers
python -m dossiergraph.cli verify-audit              # check the audit hash chain
python -m dossiergraph.cli tamper-audit              # edit one audit row directly in SQLite, then
watch verification fail
```

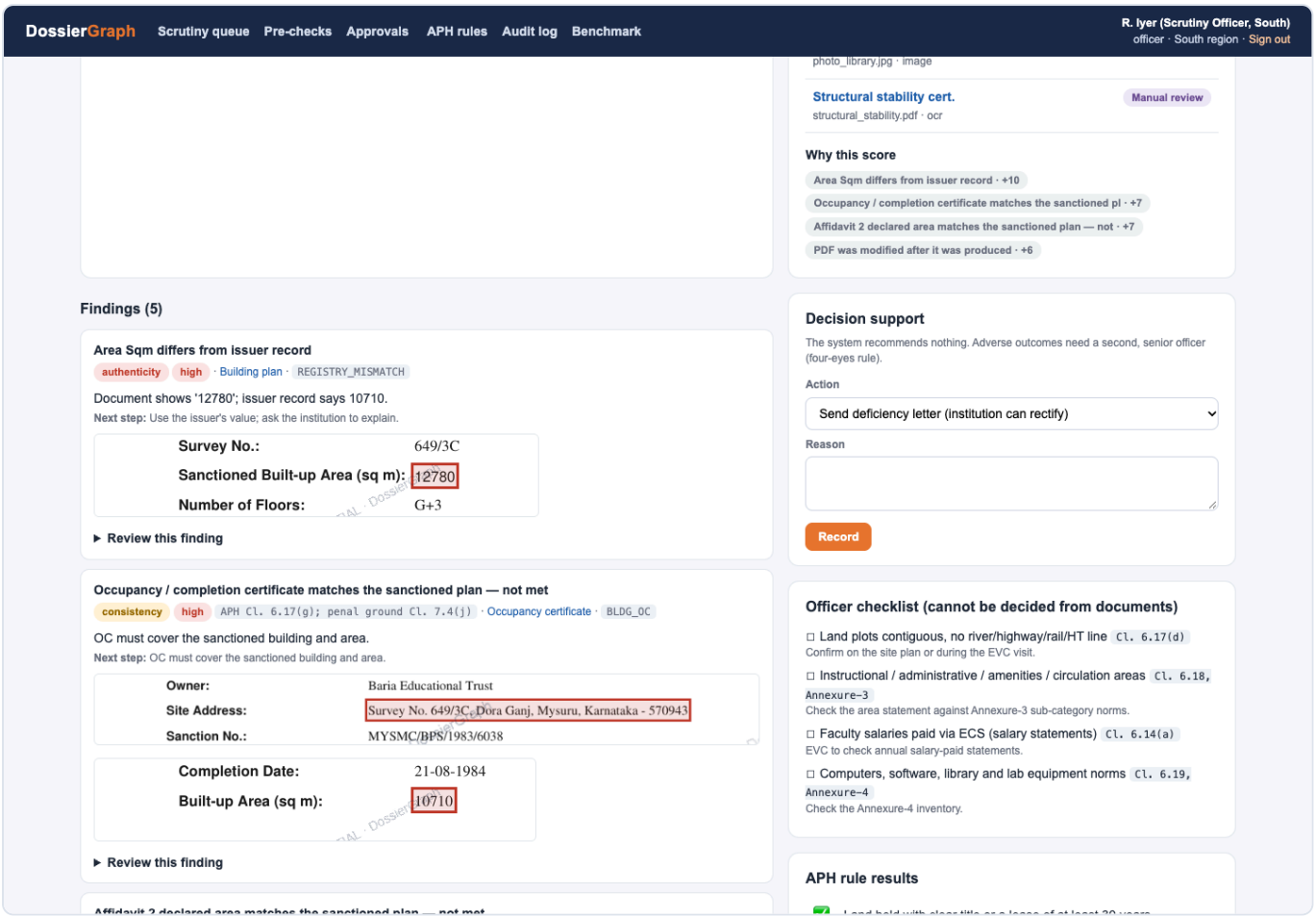
Five-minute demo

Step 1: Queue. Sign in as `officer.south`. The queue is ordered by an explainable triage score, and every score lists its top contributing factors. The officer sees the South region only.

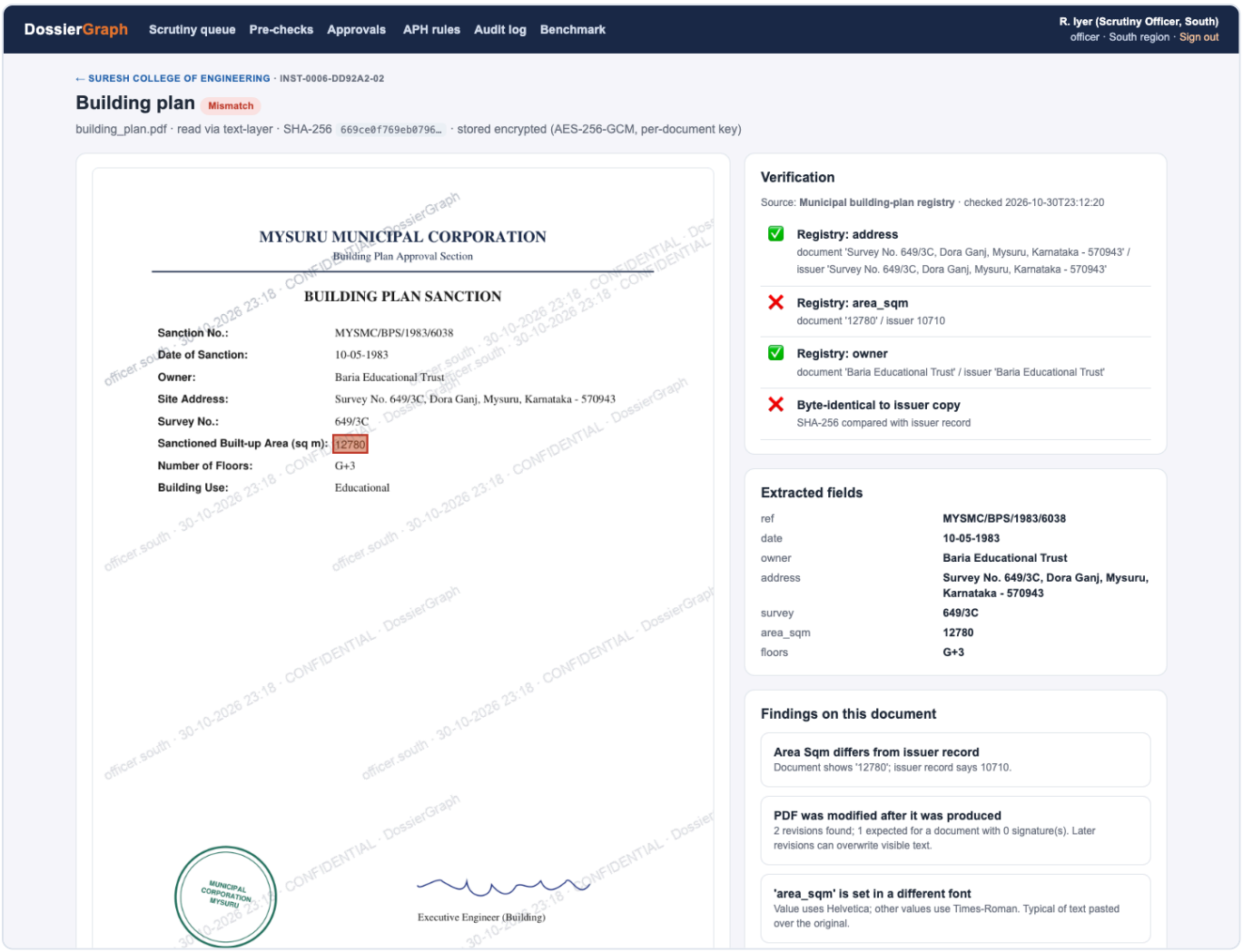


Step 2: Dossier graph. Open *Suresh College of Engineering*. The building plan is red: its sanctioned area was edited after issue. The graph shows the contradiction spreading to the occupancy certificate and Affidavit 2, which no longer agree with the plan.

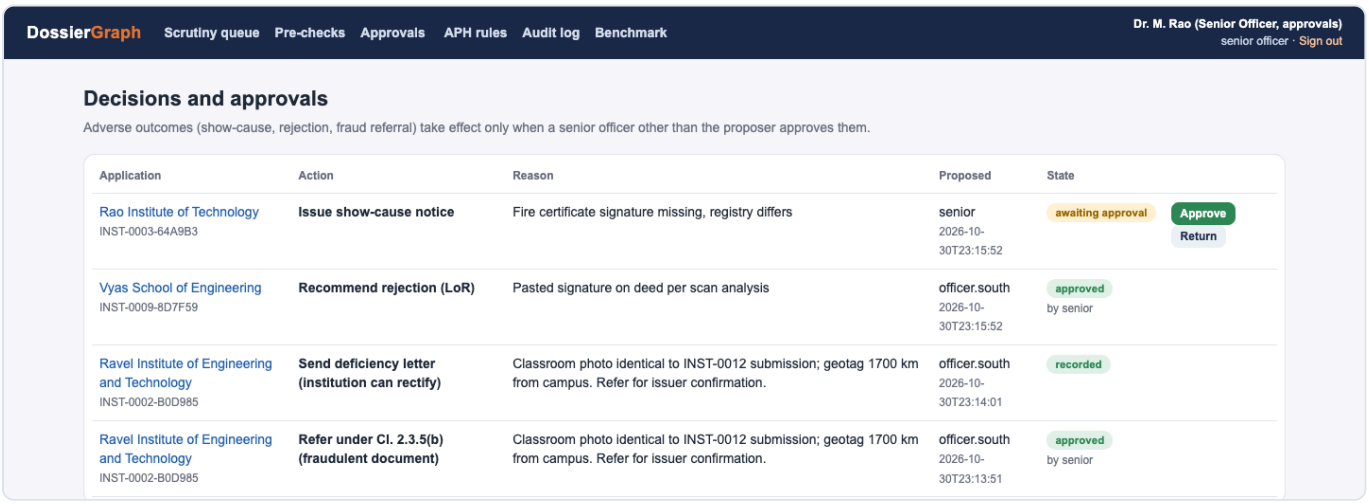
Step 3: Evidence. Scroll to the findings. Each one shows the APH clause, a watermarked crop of the exact field, and the next step. Officers accept a finding, override it with a written reason, or escalate it.



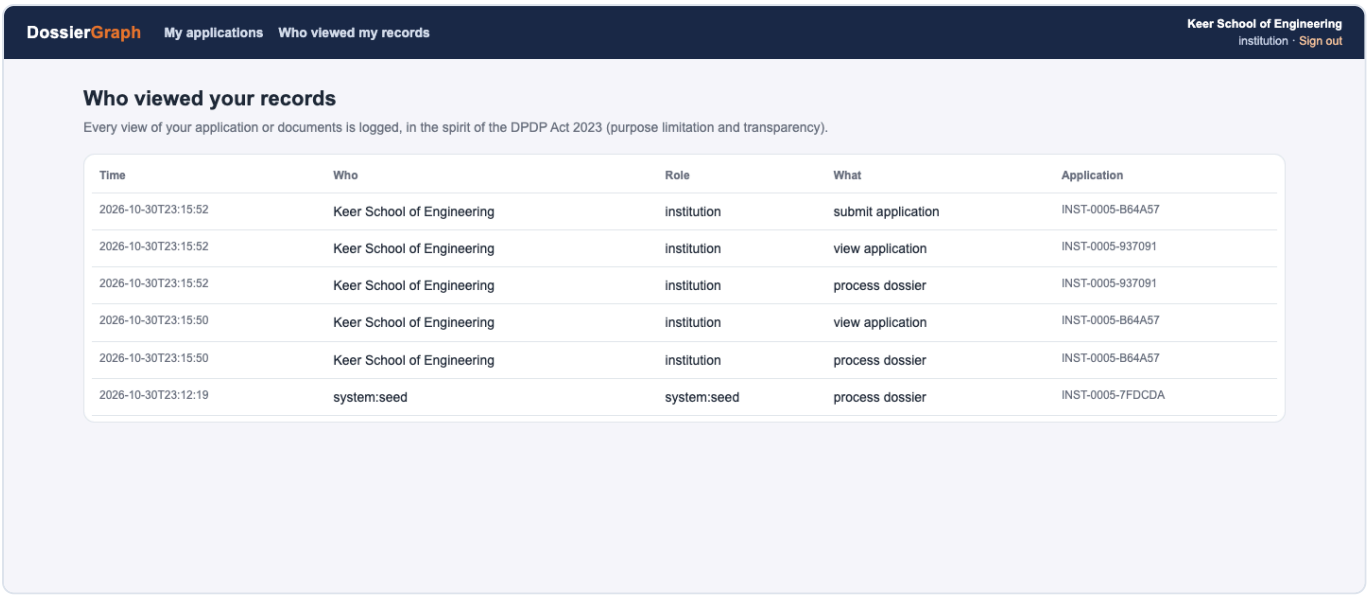
Step 4: Document view. Click the building plan. The issuer registry disagrees on the area, the PDF has an extra revision, and the edited value is set in a different font. The page is rendered on the server and watermarked with the viewer's name and the time; the original file never reaches the browser.



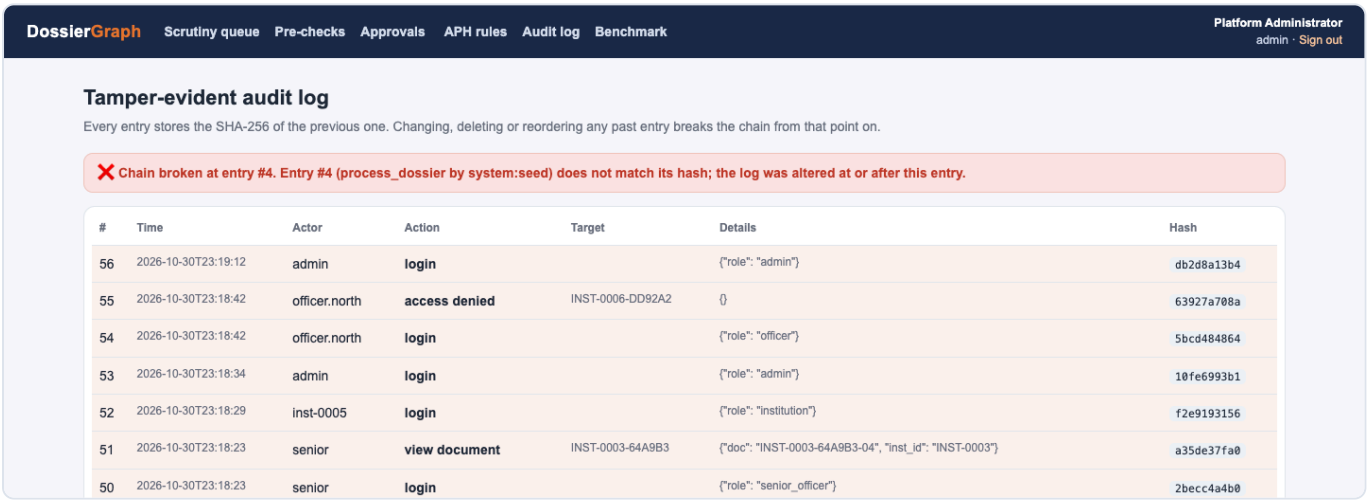
Step 5: Four-eyes rule. Propose *Refer under Cl. 2.3.5(b)* with a reason. An officer cannot approve it, and a senior officer cannot approve their own proposal. Sign in as `senior` to approve it.



Step 6: Institution side. Sign in as `inst-0005`. Run a pre-check: the same engine flags the classroom shortfall before the application is submitted. *Who viewed my records* lists every officer view of the institution's records.



Step 7: Tamper-evident log. As `admin`, open *Audit log* and see the chain verified. Run `python -m dossiergraph.cli tamper-audit` and reload: the break is pinpointed.

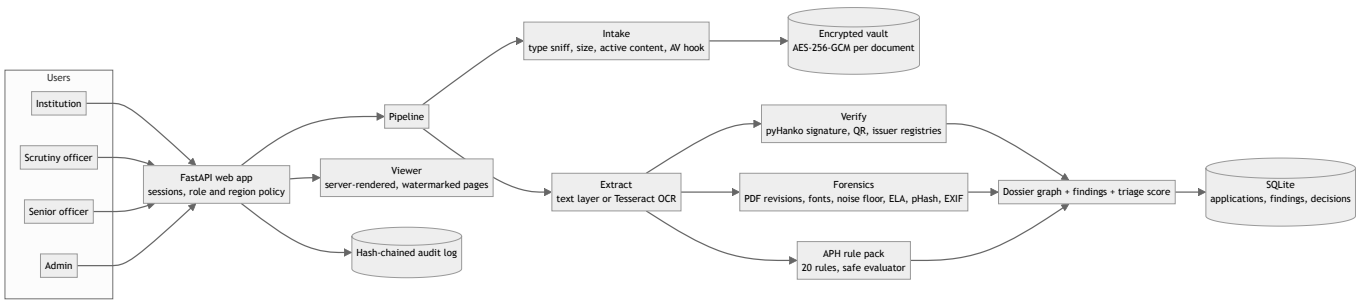


What is built

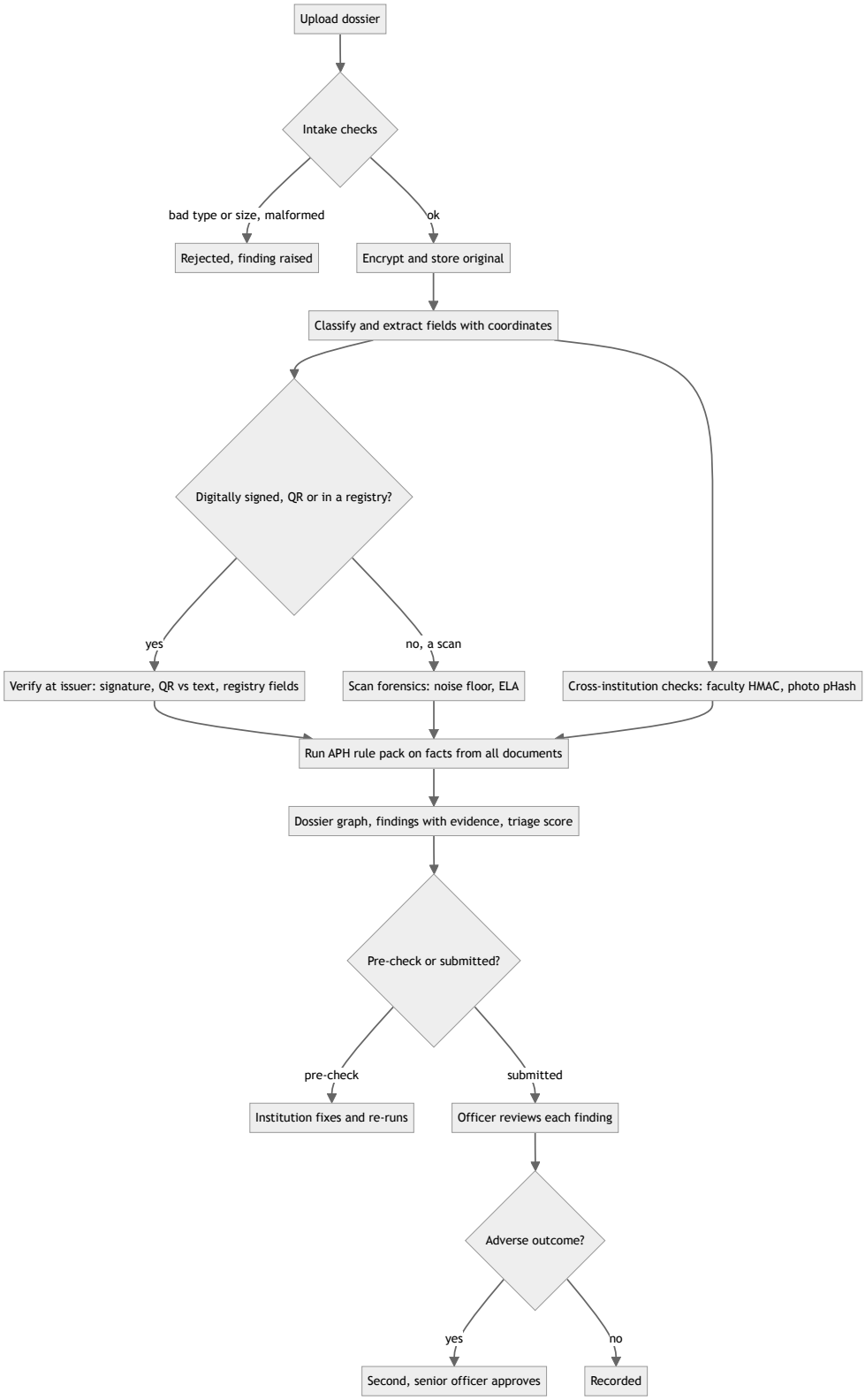
Feature (from the idea deck)	Module	Status
Category checklist and missing-document detection	`pipeline.py` (`REQUIRED`)	✅ Built
OCR + document classification + field extraction with page coordinates	`extract.py` (PyMuPDF text layer, Tesseract OCR)	✅ Built
Issuer verification: digital signatures	`verify.py` (pyHanko, test PKI in `pki.py`)	✅ Built (real PAdES validation against a test root)
Issuer verification: QR codes	`verify.py` + OpenCV QR decoder	✅ Built
Issuer verification: registries (DigiLocker-style pull, municipal, land, fire)	`verify.py` `Registry`	🟡 Mock registries with real adapter logic
6 verification statuses (verified, mismatch, expired, pending, source unavailable, manual review)	`verify.py`	✅ Built

Feature (from the idea deck)	Module	Status
Dossier cross-consistency graph	`pipeline.py` (graph) + `templates/app.html` (SVG)	✅ Built
20 APH 2024-27 rules as code with clause citations	`rules/aph_2024_27.yaml` + `rules/__init__.py`	✅ Built (15 automated, 5 officer-checklist)
PDF forensics: extra revisions, font inconsistency, edits after signing	`forensics.py`, `verify.py`	✅ Built
Scan forensics: pasted seal or signature (noise floor + ELA)	`forensics.py`	✅ Built
Photo reuse across institutions (perceptual hash) and EXIF GPS vs campus	`forensics.py`, `pipeline.py`	✅ Built
Ghost-faculty detection via keyed-hash (HMAC) matching across institutions	`crypto.py`, `pipeline.py`	✅ Built
Officer workstation: queue, explainable score, evidence crops, accept / override / escalate	`app.py`, templates	✅ Built
Four-eyes approval for adverse outcomes	`access.py`, `app.py`	✅ Built
Institution pre-check portal and submission	`app.py` `/portal`	✅ Built
Tamper-evident audit log	`db.py`	✅ Built
Envelope encryption per document; secure watermarked viewing	`crypto.py`, `viewer.py`	✅ Built
Role- and region-based access; DPDP-style access log for institutions	`access.py`, `/portal/access-log`	✅ Built
Upload hardening: type sniffing, size limit, active-content detection, antivirus hook	`intake.py`	✅ Built (ClamAV is used if installed)
Synthetic forgery benchmark with measured precision and recall	`synth/`, `bench/`	✅ Built
No-code rule builder with dry-run on past applications	—	⏭ Roadmap (rules are edited as YAML today)
Hindi / regional-language OCR	—	⏭ Needs Tesseract `hin` data or PaddleOCR
Remote EVC capture app, triage of inspection visits	—	⏭ Roadmap (the "verify on site" list is part of the decision step)

Architecture



How a dossier is processed



Every step writes to the hash-chained audit log.

Test documents

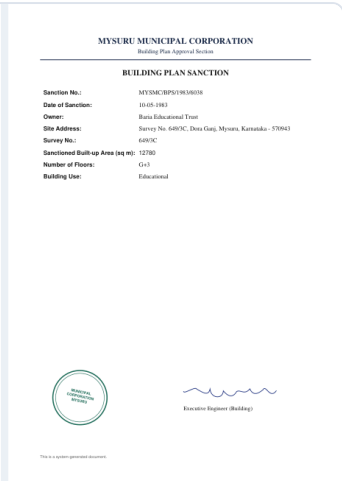
The problem statement asks for **authorized, anonymised or synthetic datasets**, and real institutional dossiers contain personal data covered by the DPDP Act. So the prototype ships a generator (`synth/`) that produces

realistic **synthetic** dossiers; every person and institution is fictitious. The demo set lives in `prototype/data/samples/demo/` (16 institutions, about 13 files each):


OFFICE OF THE SUB-REGISTRAR, CHENNAI
Department of Stamps and Registration, Government of Tamil Nadu
REGISTERED SALE DEED
Registration No.: SR/CHE/1985/52777
Date of Registration: 26-02-1985
Vendor / Lessor: Charvi Kadukia
Purchaser / Lessee: Sara Educational Trust
Survey No.: 447/4C
Locality: Chennai
District: Chennai
Land Area (acres): 11.10
Deed Type: Sale
Lease Term (years): Not applicable
The above property is conveyed free from all encumbrances and registered under the Registration Act, 1908. Book I, Volume and page as per office record.
Sub-Registrar
Chennai
Scanned sale deed (pasted signature)


INDIA NON JUDICIAL
ONE HUNDRED RUPEES
e-Stamp Certificate No. IN-TE27730021779
AFFIDAVIT 2
I, the undersigned, Chairman of the Trust, do hereby solemnly affirm that the built-up area of the institution remains unchanged and the details below are true.
Deponent: Samarth Bail
Institution: Bains School of Engineering
Address: Survey No. 61/2E, Thakkar Marg, Hyderabad, Telangana - 500082
Declared Built-up Area (sq m): 25040
Date of Affidavit: 17-10-2026
Samarth Bail
Deponent
Affidavit 2 on e-stamp paper (swapped seal)

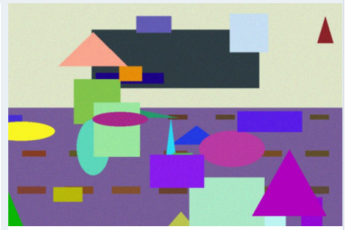

MYSURU TOWN AND COUNTRY PLANNING DEPARTMENT
Government of Karnataka
LAND USE CERTIFICATE
Certificate No.: TCP/MYS/LUC/2017/3455
Date of Issue: 14-03-2010
Applicant: Khanna Educational Trust
Survey No.: 467/9B
Land Area (acres): 17.30
Permitted Use: Educational (Institutional)
Joint Director (Town Planning)
Land-use certificate with QR (altered survey no.)


MYSURU MUNICIPAL CORPORATION
Building Plan Approval Section
BUILDING PLAN SANCTION
Sanction No.: MYSMC/BPS/1983/0038
Date of Sanction: 10-05-1983
Owner: Barra Educational Trust
Site Address: Survey No. 649/3C, Dora Ganj, Mysuru, Karnataka - 570943
Survey No.: 649/3C
Sanctioned Built-up Area (sq m): 12780
Number of Floors: G+3
Building Use: Educational
Executive Engineer (Building)
Building plan (edited after issue)


HYDERABAD MUNICIPAL CORPORATION
Occupancy Certificate Section
OCCUPANCY CERTIFICATE
Certificate No.: HYDMC/OC/1985/957
Date of Issue: 13-12-1985
Owner: Sheriff Educational Trust
Site Address: Survey No. 251/5D, Trivedi Chowk, Hyderabad, Telangana - 500104
Sanction No.: HYDMC/BPS/1984/108
Completion Date: 03-11-1985
Built-up Area (sq m): 19880
Commissioner
Occupancy certificate (digitally signed)


TELANGANA FIRE AND EMERGENCY SERVICES
Fire Prevention Wing
FIRE SAFETY CERTIFICATE
Certificate No.: TEFES/2025/02091
Date of Issue: 01-10-2026
Occupant: Sethi Institute of Technology
Premises Address: Survey No. 251/5D, Trivedi Chowk, Hyderabad, Telangana - 500104
Occupancy Type: Educational
Valid From: 01-10-2026
Valid Until: 30-09-2029
Chief Fire Officer
Fire safety certificate (digitally signed)


SETHI INSTITUTE OF TECHNOLOGY
Survey No. 251/5D, Trivedi Chowk, Hyderabad, Telangana - 500104
FACULTY ROSTER 2027-28
Institution: Sethi Institute of Technology
AICTE Permanent ID: 1-4255542466
Total Faculty: 86
S.No	Faculty ID	Name	Designation	Qualification	Joined
1	1-4036630734	Nakul Aaron	Associate Professor	M.Tech	01-01-2005
2	1-6710146638	Dr. Daniel Bunk	Assistant Professor	Ph.D.	01-11-2010
3	1-6104787722	Wyatt Balas	Assistant Professor	M.Tech	01-04-2021
4	1-2648581589	Ila Sastry	Associate Professor	M.Tech	01-07-2017
5	1-5296989831	Zilmil Kaur	Associate Professor	M.Tech	01-05-2023
6	1-1904993522	Dr. Kelah Agrawal	Professor	Ph.D.	01-10-2017
7	1-1430925927	Tanvi Mall	Assistant Professor	M.Tech	01-04-2009
8	1-2195259064	Dr. Fiyaz Kumar	Assistant Professor	Ph.D.	01-06-2024
9	1-2650587581	Victor Varghese	Assistant Professor	M.E.	01-02-2007
10	1-2041732568	Ayushman Boase	Associate Professor	M.Tech	01-06-2014
11	1-7963577895	Nandini Chokshi	Assistant Professor	M.E.	01-10-2009
12	1-2650274117	Dr. Libni Bhavsar	Assistant Professor	Ph.D.	01-03-2025
13	1-4901650223	Aadi Kohl	Assistant Professor	M.E.	01-04-2024
14	1-3379116370	Charles Singhal	Associate Professor	M.Tech	01-08-2017
15	1-5632112831	Mehbuba Sami	Assistant Professor	M.Tech	01-02-2006
16	1-9756134981	Dr. Rushil Madan	Professor	Ph.D.	01-04-2013
17	1-4398693998	Dr. Gagan Mukhopadhyay	Assistant Professor	Ph.D.	01-08-2024
18	1-3233758744	Alexander Rout	Assistant Professor	M.E.	01-03-2007
19	1-5613396482	Krish Koshy	Associate Professor	M.Tech	01-04-2011
20	1-5501882681	Dr. Advay Bhasin	Professor	Ph.D.	01-08-2018
21	1-1259654723	Vidhi Goel	Associate Professor	M.Tech	01-05-2010
22	1-5179253525	Sara Upadhyay	Assistant Professor	M.E.	01-06-2007
23	1-9121530442	Karan Saini	Associate Professor	M.Tech	01-11-2015
24	1-5731822290	Jackson Rana	Assistant Professor	M.Tech	01-12-2018
25	1-5135073017	Simon Acharya	Assistant Professor	M.E.	01-10-2005
26	1-3071186080	Dr. Diksha Bahri	Professor	Ph.D.	01-06-2014
27	1-5292208544	Rachit Ramesh	Associate Professor	M.Tech	01-08-2007
28	1-6782188561	Brijesh Dhingra	Assistant Professor	M.Tech	01-11-2007
29	1-4869869923	Dr. Riya Kunvarila	Assistant Professor	Ph.D.	01-01-2018
30	1-3460189480	Dr. Falguni Mathur	Assistant Professor	Ph.D.	01-11-2024
31	1-9643394787	Dr. Ishan Borde	Assistant Professor	Ph.D.	01-02-2007
32	1-7340220611	Dr. Subodh Kibe	Professor	Ph.D.	01-07-2013
Faculty roster (multi-page)


Geotagged campus photo

Document	Format	How it is verified
Registered sale or lease deed	Scanned PDF (tinted paper, noise, slight skew)	Land-records registry (state-dependent: available, pending or unavailable); scan forensics
Land-use certificate	Digital PDF with QR code	QR payload vs printed text; Town Planning registry
Building plan sanction	Digital PDF	Municipal registry; PDF revision and font forensics
Occupancy certificate	**Digitally signed** PDF (PAdES, test CA)	Signature validation, including edits after signing; municipal registry
Fire safety certificate	**Digitally signed** PDF	Signature, expiry against the academic year, address; fire-services registry
Structural stability certificate	Scanned PDF (only for buildings older than 30 years)	Manual review; scan forensics
Affidavit 2	Scanned on e-stamp paper with notary seal	Manual review; scan forensics; area vs building plan
Faculty roster	Multi-page digital PDF	1:20 / 1:25 ratios; cross-institution HMAC matching
Campus photos (4)	JPEG with EXIF GPS	Perceptual-hash reuse; GPS vs campus
`application.json`	Portal form data	Source of declared intake, address, classrooms and GPS

Using your own documents: sign in as an institution and upload PDFs or JPEGs in *Run a pre-check*; uploads replace files of the same name. Documents from unknown issuers fall back to *manual review*. Real registries

(DigiLocker requester APIs, state portals) need authorisation from the issuing agencies; `verify.Registry` is the adapter boundary where they plug in.

The 8 forgery types and how each is caught

#	Forgery (as generated)	Main signals	Benchmark recall
1	**Altered date**: fire-certificate expiry extended, either edited after signing or retyped unsigned	Signature covers an earlier revision, or an expected signature is missing; registry mismatch	10/10
2	**Altered identifier**: land-use certificate issued for another parcel, survey number retyped	QR contradicts the printed text; registry mismatch	10/10
3	**Swapped seal**: notary seal on Affidavit 2 replaced by a pasted image	Noise-free region in the scan; ELA peak	10/10
4	**Pasted signature** on the scanned deed	Noise-free region in the scan	10/10
5	**PDF text edit**: sanctioned area raised via an incremental save	Extra PDF revision; value in a different font; registry mismatch; cross-document rules fail	10/10
6	**Photo reuse**: classroom photo copied from another institution	pHash distance ≤ 8 to the other institution's photo; GPS 100s of km from campus	10/10
7	**Faculty double-claim**: faculty IDs also on another institution's roster	HMAC token collision (raw IDs are never compared or shown)	10/10
8	**Cross-document inconsistency**: Affidavit 2 declares 15–30% more area than the plan	`BUA_AFFIDAVIT2` rule (2% tolerance)	10/10

Forensic signals are **indicators**, never proof. The UI and the audit trail say so, and the final call is always the officer's.

APH rule pack

`prototype/dossiergraph/rules/aph_2024_27.yaml`. Clause numbers were checked against the official APH 2024-25 to 2026-27 text.

Rule	Clause	Check
LAND_TITLE · LAND_LIVE_LEASE · LAND_DEED_TYPE	Cl. 6.17(a), (b)	Sale deed or lease ≥ 30 years; remaining lease ≥ longest programme; registered deed
LAND_USE_CERT	Cl. 6.17(e)	Land-use certificate names the deed's survey number
BLDG_PLAN_MATCH · BLDG_OC	Cl. 6.17(f), (g); 7.4(j)	Plan is for this site; OC matches plan address and area
BLDG_STRUCT_STAB	Cl. 6.17(h)	Building > 30 years old ⇒ stability certificate ≤ 5 years old
BLDG_FIRE	Cl. 6.17(i)	Fire certificate valid on 1 July 2027 and issued for the plan's address
AUTHORITY_VALID	Cl. 6.17(j), Annex-14	Issuer is a competent authority (list mocked)
BUA_CLASSROOMS · BUA_AFFIDAVIT2	Annex-3; Affidavit 2, Cl. 6.39	One classroom ≥ 66 m² per division-year; declared area within 2% of plan
FAC_RATIO_UG · FAC_RATIO_DIP · FAC_ID_UPDATE	Cl. 6.14, Annex-5; 6.14(c)	1:20 UG, 1:25 diploma; no faculty claimed elsewhere; roster count consistent

Rule	Clause	Check
FRAUD_ESCALATE	Cl. 2.3.5(b)	No unresolved high-severity authenticity indicators
5 officer-checklist rules	6.17(d), 6.18, 6.14(a), 6.19, Joint Affidavit 7	Cannot be decided from documents, so listed for the officer / EVC

Rules use a small expression language parsed with Python's `ast`, with a whitelist of functions. Nothing is passed to `eval`, and private attributes are rejected (both covered by tests). If a fact is missing, the rule is marked **undetermined** and sent to the officer; it never silently passes. Each application stores the rule-pack version it was checked against.

Trust layer (security and privacy)

Control	Implementation in the prototype
Encryption at rest	Each document has its own random AES-256-GCM key, wrapped by a master key (`data/keys/master.key`). Ciphertext is bound to the document ID, so swapping files fails to decrypt.
Secure viewing	Originals are never sent to the browser. Pages are rasterised on the server (which disarms embedded JavaScript and actions), outlined where there is evidence, and watermarked with the viewer and time. Responses are `no-store`.
Access control	Session login (scrypt password hashes). Policy in `access.py`: institutions see their own records only, officers their region only, admins cannot decide. Denials are audited.
Four-eyes rule	Show-cause, rejection and fraud referral take effect only after a *different* senior officer approves. Enforced on the server and covered by tests.
Privacy-preserving matching	Faculty IDs are compared as HMAC-SHA256 tokens under a secret key. Institutions are told only "another institution", never which one.
Tamper-evident audit log	Each entry hashes the previous one; `verify_audit` pinpoints the first altered entry. Logins, views, reviews, decisions and denials are all logged.
DPDP transparency	Institutions see who viewed their records and when.
Upload hardening	Magic-byte type check, 15 MB limit, malformed-PDF rejection, detection of `/JavaScript`, `/OpenAction`, `/Launch`, `/EmbeddedFiles` and similar, and an optional ClamAV scan.
No third-party AI	All processing is local: Tesseract, OpenCV and pyHanko. No document leaves the machine.
Human oversight	Findings are labelled as indicators. Overrides require a reason. The system never issues an outcome.

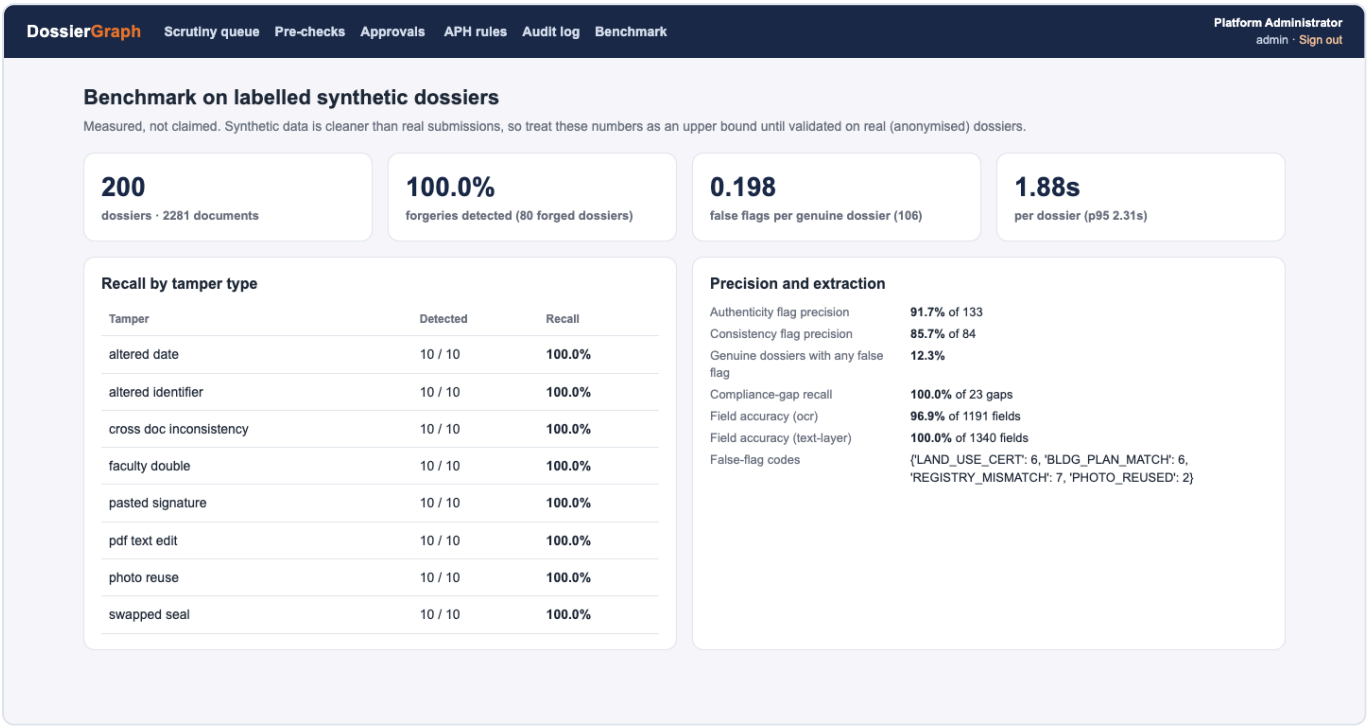
Benchmark results

`python -m bench.run_benchmark --n 200 --seed 2026`. Results are in `prototype/bench/results.json` and on the in-app *Benchmark* page.

- **Data:** 200 synthetic dossiers, 2,281 documents. 80 contain one forgery (10 of each type), 23 have genuine compliance gaps, and 106 are fully genuine.
- **Method:** thresholds were tuned on *different* seeds (5, 9 and 77), and the benchmark seed was never used for tuning.

Metric	Result
Forgeries detected (recall)	**80 / 80 (100%)**, 10/10 for every type
Authenticity-flag precision	**91.7%** (133 flags)

Metric	Result
Consistency-flag precision	**85.7%** (84 flags)
False flags per genuine dossier	**0.198**; 12.3% of genuine dossiers had at least one
Compliance-gap recall	**100%** (23 / 23)
Field extraction accuracy	**100%** text layer (1,340 fields) · **96.9%** OCR (1,191 fields)
Processing time	**1.9 s** mean per dossier, 2.3 s p95 (single laptop process)



Where it went wrong (all false flags on genuine dossiers):

13 of the 106 genuine dossiers got at least one false flag:

- **11 were OCR misreads on scanned land deeds**, for example `7.35` read as `1.35`, or a `/` dropped from a survey number. A misread deed survey number then also fails the two cross-document land rules. One-character slips are already downgraded to "possible OCR misread"; larger misreads are not.
- **2 were one pair of genuine building photos** at a perceptual-hash distance of exactly 8, the threshold.

Read these numbers carefully. Synthetic forgeries are cleaner and more uniform than real ones, so 100% recall here is an **upper bound**, not a promise. Real scans will be noisier, so expect lower OCR accuracy and more false flags. The next step is validation on authorised, anonymised real dossiers.

Tests

`python -m pytest -q` runs 14 tests (about 30 s) on an isolated data folder:

- **Encryption:** encryption round-trip, tamper detection, and the ciphertext's binding to its document ID.
- **Keyed hashing:** HMAC faculty tokens are normalised and never contain the raw ID.
- **Audit log:** the chain detects an edited row and pinpoints it.
- **Access policy:** role and region rules; the four-eyes rule blocks an officer and a self-approving senior.
- **Passwords:** scrypt password hashing.
- **Intake:** rejects a disguised executable and detects a JavaScript `/OpenAction`.

- **Rule language:** rejects `__import__`, lambdas, `open()` and private attributes; evaluates rules; marks a missing fact as *undetermined*.
- **End to end:** one dossier per forgery type is generated and processed, and every type is caught. The audit chain stays intact and stored documents are encrypted.

Prototype vs production

Concern	Prototype	Production target (from the idea deck)
Web UI	Server-rendered FastAPI + Jinja (no build step)	Next.js workstation
Database	SQLite	PostgreSQL
Object store	Encrypted files in `data/vault/`	MinIO (S3-compatible), on government infrastructure
Keys	Local master key file	HashiCorp Vault transit (key rotation, HSM)
Identity	Built-in login and policy functions	Keycloak (MFA) + OPA policies
Background jobs	In-process	Celery + Redis workers
OCR	Tesseract (English)	PaddleOCR / Tesseract with Hindi and regional models
Issuer sources	Mock registries; real signature and QR checks	DigiLocker requester APIs, state portals (needs MoUs)
PKI	Self-made **TEST** root CA	CCA-licensed certificate chains
Antivirus	Optional ClamAV	ClamAV / enterprise AV

Known limitations and next steps

1. **OCR on real scans.** Add 300-dpi re-scan guidance, image clean-up and an OCR ensemble. Treat low-confidence numeric fields as *manual review* rather than mismatch.
2. **Perceptual hashing alone** can collide on similar buildings. Add a second check (ORB keypoint matching) before raising a high-severity reuse flag.
3. **Forgery realism.** Add harder forgeries (copy-move inside one scan, full re-typesetting with matching fonts, printed-and-rescanned edits) and validate on real anonymised dossiers.
4. **No-code rule builder** with a dry run on past applications ("this change would flag 37 more"), and approval of rule changes by a second admin.
5. **Annexure-3 / Annexure-14 tables:** the built-up-area sub-categories and the state-wise competent authorities still need to be transcribed in full.
6. **Real connectors:** DigiLocker requester (OAuth + PKCE, India-hosted), state fire and land portals, and the AICTE faculty registry.
7. **Hardening:** CSRF tokens, rate limiting, HTTPS, key rotation, and data-retention jobs.

Repository layout

```
SIH 2026/
├── README.md                     this file (PDF copy: docs/DossierGraph_README.pdf)
├── SIH26261_Feature_Plan.md/.pdf  feature plan and idea-submission plan
├── Deck/                          idea-submission decks (PPTX + PDF)
├── reports/, research_notes/      SIH research and evaluation
├── docs/screenshots/              screenshots used in this README
└── prototype/
    ├── dossiergraph/              the application
    │   ├── app.py                 web routes (portal, queue, workstation, approvals, audit)
    │   ├── pipeline.py            end-to-end dossier processing, graph, score
    │   ├── extract.py             classification, text layer, OCR, field extraction
    │   ├── verify.py              signatures, QR, issuer registries, statuses
    │   ├── forensics.py           PDF, scan and photo forensics
    │   ├── rules/                 APH rule pack (YAML) + safe evaluator
    │   ├── crypto.py · access.py · db.py · intake.py · viewer.py · pki.py
    │   └── templates/, static/
    ├── synth/                     synthetic document and forgery generator
    ├── bench/                     benchmark runner + results.json
    ├── tests/                     pytest suite
    ├── tools/screenshots.py       regenerates README screenshots
    ├── seed_users.yaml            TEST accounts (local only)
    └── data/samples/demo/         16 demo dossiers (keys, vault and databases are not committed)
```

Team work split

Track	Owns	Start here
OCR and extraction	Better OCR, Hindi, field schemas	`extract.py`, benchmark field accuracy
Rules engine	Annexure tables, rule builder, dry run	`rules/`, `pipeline.py`
Forensics and computer vision	ORB photo matching, copy-move, harder forgeries	`forensics.py`, `synth/`
Verification adapters	DigiLocker / state portal adapters	`verify.py` `Registry`
Trust layer	Keycloak/OPA, Vault, CSRF, retention	`access.py`, `crypto.py`, `db.py`
Frontend and demo	Next.js port, demo video, deck screenshots	`templates/`, `tools/screenshots.py`

All outputs are decision support. Statutory scrutiny, inspections and final regulatory decisions remain with the competent AICTE authorities. All data in this repository is synthetic.